PRACTICE SHEET

1. How many 3–digit numbers, each less than 600, can be formed from {1, 2, 3, 4, 7, 9} if repetition of digits is allowed?
(a) 216 (b) 180
(c) 144 (d) 120

2. There are four chairs with two chairs in each row. In how many ways can four persons be seated on the chairs, so that no chair remains unoccupied?
(a) 6 (b) 12
(c) 24 (d) 48

3. In how many ways can the letters of the world CORPORATION be arranged so that vowels always occupy even places?
(a) 120 (b) 2700
(c) 720 (d) 7200

4. In fall permutations of the letters of the 'LAGAN' are arranged as in dictionary, then what is the rank of 'NAAGL'?
(a) 48^{th} Word (b) 49^{th} Word
(c) 50^{th} Word (d) 51^{st} Word

5. If a secretary and a joint secretary are to be selected from a committee of 11 members, then in how many ways can they be selected?
(a) 110 (b) 55
(c) 22 (d) 11

6. In how many ways can 7 persons stand in the form of a ring?
(a) P(7, 2) (b) 7!
(c) 6! (d) 7!/2

7. On a railway route there are 20 stations. What is the number of different tickets required in order that it may be possible to travel from every station to every other station?
(a) 40 (b) 380
(c) 400 (d) 420

8. What is the number of five – digit numbers formed with 0, 1, 2, 3, 4 without any repetition of digits?
(a) 24 (b) 48
(c) 96 (d) 120

9. A group consists of 5 men and 5 women. If the number of different five–person committees containing k men and (5–k) women is 100, what is the value of k?
(a) 2 only (b) 3 only
(c) 2 or 3 (d) 4

10. If 7 points out of 12 are in the same straight line, then what is the number of triangles formed?
(a) 84 (b) 175
(c) 185 (d) 201

11. In how many ways can 3 books on Hindi and 3 books on English be arranged in a row on a shelf, so that not all the Hindi Books are together?
(a) 144 (b) 360
(c) 576 (d) 720

12. How many words, with or without meaning can be formed by using all the letters of the word 'MACHINE' so that the vowels occurs only the odd positions?
(a) 1440 (b) 720
(c) 640 (d) 576

13. From 7 men and 4 women a committee of 6 is to be formed such that the committee contains at least two women. What is the number of ways to do this?
(a) 210 (b) 371
(c) 462 (d) 5544

14. If $P(32,6) = k\,C\,(32, 6)$, then what is the value of k?
(a) 6 (b) 32
(c) 120 (d) 720

15. What is the smallest natural number n such that $n!$ is divisible by 990?
(a) 9 (b) 11
(c) 33 (d) 99

16. What is the value of r, if $P\,(5, r) = P\,(6, r-1)$?
(a) 9 (b) 5
(c) 4 (d) 2

17. What is the number of words formed from the letters of the word 'JOKE' so that the vowels and consonants alternate?
(a) 4 (b) 8
(c) 12 (d) None of these

18. If $C\,(n, 12) = C\,(n, 8)$, then what is the value of $C(22, n)$?
(a) 131 (b) 231
(c) 256 (d) 292

19. In a football championship 153 matches were played. Every team played one match with each other team. How many teams participated in the championship?
(a) 21 (b) 18
(c) 17 (d) 15

20. How many times does the digit 3 appear while writing the integers from1 to 1000?
(a) 269 (b) 308
(c) 300 (d) None of these

21. In how many ways can a committee consisting of 3 men and 2 women be formed from 7 men and 5 women?
(a) 45 (b) 350
(c) 700 (d) 4200

22. What is the number of signals that can be sent by 6 flags of different colours taking one or more at a time?
(a) 21 (b) 6
(c) 720 (d) 1956

23. What is the number of words that can be formed from the letters of the word 'UNIVERSAL', the vowels remaining always together?
(a) 720 (b) 1440
(c) 17280 (d) 21540

24. A team of 8 players is to be chosen from a group of 12 players. Out of the eight players one is to be elected as captain and another vice-captain. In how many ways can this be done?
(a) 27720 (b) 13860
(c) 6930 (d) 495

25. What is the number of three-digit odd numbers formed by using the digits 1, 2, 3, 4, 5, 6 if repetition of digits is allowed? :
(a) 60 (b) 108
(c) 120 (d) 216

26. What is the number of ways of arranging the letters of the word 'BANANA' so that no two N's appear together?
(a) 40 (b) 60
(c) 80 (d) 100

ANSWER KEYS

1.	c	**2.**	c	**3.**	d	**4.**	b	**5.**	a	**6.**	c	**7.**	b	**8.**	c	**9.**	c	**10.**	c
11.	c	**12.**	d	**13.**	b	**14.**	d	**15.**	b	**16.**	c	**17.**	b	**18.**	b	**19.**	b	**20.**	c
21.	b	**22.**	d	**23.**	c	**24.**	a	**25.**	b	**26.**	a								

Solutions

Sol.1. (c)
There digit number less than 600 will have first element 100, and last element 599. First place will not have digit more than 6, hence, 7 and 9 cannot be taken: So first digit can be selected in 4 ways. Second digit can be selected in 6 ways and since repetitions of digits are allowed, third digit can also be selected in 6 ways: So, number of ways are 4 × 6 × 6 = 144.

Sol.2. (c)
First chair can be occupied in 4 ways and second chair can be occupied in 3 ways, third chair can be occupied in 2 ways and last chair can be occupied in one ways only. So total number of ways = 4 × 3 × 2 × 1 = 24

Sol.3. (d)
CORPORATION is 11 letter word.
It has 5 vowels (O, O, O, A, I) and 6 consonants (C, R, P, R, T, N)
In 11 letters, there are 5 even places (2nd, 4th, 6th, 8th and 10th positions)
5 vowels can take 5 even places in $\frac{5!}{3!}$ ways
(∵ Since O is repeated thrice)
Similarly, 6 consonants can take 6 odd places in $\frac{6!}{2!}$ ways
(∵ R is repeated twice)
∴ Total number of ways = $\frac{5!}{3!} \times \frac{6!}{2!}$ = 20 × 360
= 7200

Sol.4. (b)
Starting with the letter A and arranging the other four letters, there are 24 words. There are the first 24 words. Then starting with G that comes next in dictionary order and arranging A, A, L, N in different ways, there are $\frac{4!}{2!}$ = 12 words. Next the 37th word starts with L that comes next in dictionary order there are 12 words starting with L. This accounts up to the 48 words. The 49th word is 'NAAGL'

Sol.5. (a)
Selection of 2 members out of 11 has ${}^{11}C_2$
number of ways of arrangement = 55 × 2 = 110

Sol.6. (c)
Number of ways in which 7 person can stand in the form of ring = (7 –1)! = 6!

Sol.7. (b)
From each railway station, there are 19 different tickets to be issued. There are 20 railway station. So, total number of tickets = 20× 19 = 380

Sol.8. (c)
To make a 5 digit number, 0 cannot come in the beginning. So, it can be filled in 4 ways. Rest of the places can be filled in 4! Ways. So total number of digit formed = 4 × 4! = 4 × 24 = 96

Sol.9. (c)
K men selected out of 5 and 5 – k women out of 5. These are ${}^5C_k$ and ${}^5C_{5-k}$
According to problem:
${}^5C_{1C} \times {}^5C_{5-1C} = 100$

$$\Rightarrow \frac{5!}{k!(5-k)!} \times \frac{5!}{(5-k)!5!} = 100$$

$$\Rightarrow \left(\frac{5}{k!(5-k)!}\right)^2 = 100$$

$$\Rightarrow \frac{5!}{k!(5-k)!} = 10$$

This is true of k = 2 or 3.

Sol.10. (c)
Number of triangles formed from 12 point = ${}^{12}C_3$
Since 7 pars are collinear, then ${}^7C_3$ triangle will not be formed so.
$= {}^{12}C_3 - {}^7C_3$

$$= \frac{12!}{3!9!} - \frac{7!}{3!4!} = \frac{12.11.10}{3.2.1} - \frac{7.6.5}{3.2.1}$$

=220–35=185

Sol.11. (c)
Total number of arrangement = 6! = 720
Total number of arrangement while all the Hindi books are together = 4!×3!= 24 × 6 = 144
∴ The number of ways, in which books are arranged, while all the Hindi books are not together
= 720 – 144 = 576

Sol.12. (d)
There are three vowels and they have four odd places to arrange. Other letters are four and has four places to arrange.
∴ The number of words = ${}^4P_3 \times 4!$

$$= \frac{4!}{(4-3)!} \times 4! = 576$$

Sol.13. (b)
The required number of ways
$= {}^{11}C_6 - ({}^7C_6 \times {}^4C_0 + {}^7C_5 \times {}^4C_1)$

$$= \frac{11 \times 10 \times 9 \times 8 \times 7}{5 \times 4 \times 3 \times 2} - \left(7 + \frac{7 \times 6}{2} \times 4\right)$$

=462 – (7 + 84) = 371

Sol.14. (d)
Since ${}^{33}P_6 = k\ {}^{32}C_6$

$$\Rightarrow \frac{32!}{(32-6)!} = k.\frac{32!}{6!(32-6)!}$$

$\Rightarrow k = 6! = 720$

Sol.15. (b)
Consider option 'a'
Let us take $n = 9$
Since, 9! = 9×8×7×6×5×4×3×2×1=362880
Which is not divisible by 990.
Now assume, $n = 11$
Since 11! = 39916800
Which is divisible by 990.
Thus, required smallest natural number 11

Sol.16. (c)
Given P (5, r) = P (6, r –1)
$\Rightarrow {}^5P_r = {}^6P^{r-1}$

$\Rightarrow \frac{5!}{(5-r)!} = \frac{6!}{(6-r+1)!}$

$\Rightarrow \frac{5!}{(5-r)!} = \frac{6!}{(7-r)!}$

$\Rightarrow \frac{5!}{(7-r)!} = \frac{6\times5!}{(7-r)(6-r)(5-r)!}$

$\Rightarrow(7-r)(6-r) = 6$

$\Rightarrow 42-13r+r^2 = 6$

$\Rightarrow r^2 - 13r + 36 = 0$

$\Rightarrow r^2 - 9r - 4r + 36 = 0$

$\Rightarrow (r-9)(r-4)=0$

$\Rightarrow r = 4$ ($\because r\neq9$)

Sol.17. (b)
Total number of letters = 4
No. of vowels = 2
No. of consonants = 2
Possibilities of words formed from the letters of word "JOKE" are
JOKE, KOJE, KEJO, JEKO, EJOK, EKOJ, OKEJ, OJEK
Thus, required number of words = 8

Sol.18. (b)
Given $C(n, 12) = C(n, 8)$
$\Rightarrow {}^nC_{12} = {}^nC_8$

$\Rightarrow \frac{n!}{(n-12)!12!} = \frac{n!}{(n-8)!8!}$

$\Rightarrow \frac{1}{(n-12)!(12\times11\times10\times9\times8!)}$

$\Rightarrow \frac{1}{(n-8)(n-9)(n-10)(n-11)(n-12)!8!}$

$\Rightarrow \frac{1}{12\times11\times10\times9} = \frac{1}{(n-8)(n-9)(n-10)(n-11)}$

$\Rightarrow(n-8)(n-9)(n-10)(n-11)$
$=12 \times 11 \times 10 \times 9$

$\Rightarrow n-8=12$, $n-9=11$, $n-10=10$ and $n-11=9$
$\Rightarrow n=20$
$\Rightarrow C(22, n) = {}^{22}C_{20}$

$= \frac{22!}{2!20!} = \frac{22\times21}{2} = 231$

Sol.19. (b)
Let total no. of team participated in a championship be *n*. Since every team played one match with each other team.

$\therefore {}^nC_2=153 \Rightarrow \frac{n!}{2!(n-2)!} = 153$

$\Rightarrow \frac{n(n-1)(n-2)!}{2!(n-2)!} = 153 \Rightarrow \frac{n(n-1)}{2} = 153$

$\Rightarrow n(n-1)=306$
$\Rightarrow n^2 - n - 306 = 0$
$\Rightarrow n(n-18) + 17(n-18) = 0$
$\Rightarrow n = 18, -17$
N cannot be negative
$\therefore n \neq -17$
$\neq n =18$

Sol.20. (c)
Before 1000 there are one digit, two digits and three digits numbers.
Number of ties 3 appear in one digit number = 20 × 9
Number of times 3 appear in two digit number = 11 × 9
Number of times 3 appear in three digit numbers = 21
Hence total number of times the digit 3 appear while writing the integers from 1 to 1000
= 180 + 99 + 21= 300

Sol.21. (b)
Total no. of Men=7
Total no. of women=5
Required number of ways= ${}^7C_3 X {}^5C_2$

$= \frac{7!}{3!4!}\times\frac{5!}{2!n!} = \frac{7\times6\times5}{3\times2}\times\frac{5\times4}{2}$

$7\times5\times10 = 35\times10 = 350$

Sol.22. (d)
Required number of ways
$={}^6P_1+{}^6P_2+{}^6P_3+{}^6P_4+{}^6P_5+{}^6P_6$
$= 6 + 30 + 120 + 360 + 720 + 720 =1956$

Sol.23. (c)
Consider the word UNIVERSAL
Total no. of vowels= U,I,E,A=4
Let us consider these as a single letter
UNIVERSAL
Then, total No. of letters =6
Then, number of ways to arrange them=6!=720.
But vowels can also arranged in 4! Or 24 ways.
Hence, total number of ways=720 X 24 =17280

Sol.24. (a)
Total no. of players=12
No. of chose players=8
Number of ways to choose 8 players from 12 players

$={}^{12}C_8 = \frac{12!}{8!4!} = \frac{12\times11\times10\times9\times8!}{8!4!} = 495$

Since, out of the 8 players 1 is to be elected as captain and another vice-captain
therefore number of ways to choose a captain and a vice-captain.

$={}^8C_1 \times {}^7C_1 = 8\times7 = 56(\because \quad)$

Here, required number of ways = 495×56 = 27720

Sol.25. (b)
Total No. of digits=6
To form a odd numbers we have only 3 choice for the unit digits.
Now, Extreme left place can be filled in 6 ways the middle.
$\therefore$ Required number of numbers = 6 x 6 x 3 = 108

Sol.26. (a)
Total no. of letters in BANANA = 6
No. of repeated letter N = 2
No. of repeated letter A =2
Therefore , Number of ways that can be formed by using the words

'BANANA' $= \frac{6!}{3!2!} = \frac{6\times5\times4\times3!}{3!\times2!} = 60$

Number of ways in which two N comes together= $\frac{5!}{3!}$=20
$\therefore$ Required number of ways
= 60 –20= 40

NDA PYQ

1. A, B, C, D and E are coplanar points and three of them lie in a straight line. What is the maximum number of triangles that can be drawn with these points as their vertices?
(a) 5 (b) 9
(c) 10 (d) 12
[NDA (I) - 2011]

2. Using the digits 1,2, 3,4 and 5 only once, how many numbers greater than 41000 can be formed?
(a) 41 (b) 48
(c) 50 (d) 55
[NDA (I) - 2011]

3. What is the value of n, if $P(15,n-1) : P(16, n-2) = 3 : 4$?
(a) 10 (b) 12
(c) 14 (d) 15
[NDA (I) - 2011]

4. In how many ways 6 girls can be seated in two chairs?
(a) 10 (b) 15
(c) 24 (d) 30
[NDA (I) - 2011]

5. 5 books are to be chosen from a lot of 10 books. If m is the number of ways of choice when one specified book is always included and n is the number of ways of choice when a specified book is always excluded, then which one of the following is correct?
(a) $m > n$ (b) $m = n$
(c) $m = n - 1$ (d) $m = n - 2$
[NDA (I) - 2011]

6. What is the total number of combination of n different things taken 1, 2, 3,….,n at a time?
(a) 2^{n+1} (b) 2^{2n+1}
(c) 2^{n-1} (d) $2^n - 1$
[NDA (I) - 2011]

7. What is the value of $\sum_{r=1}^{n} \frac{P(n,r)}{r!}$?
(a) 2^n-1 (b) 2^n
(c) 2^{n-1} (d) $2^n + 1$
[NDA (II) - 2011]

8. There are 4 candidates for the post of a lecturer in Mathematics and one is to be selected by votes of 5 men. What is the number of ways in which the votes can be given?
(a)1048 (b) 1072
(c)1024 (d) 625
[NDA (II) - 2011]

9. How many diagonals will be there in an n-sided regular polygon?
(a) $\frac{n(n-1)}{2}$ (b) $\frac{n(n-3)}{2}$
(c) n^2-n (d) $\frac{n(n+1)}{2}$
[NDA (II)-2011]

10. What is the number of ways that 4 boys and 3 girls can be seated so that boys and girls alternate?
(a) 12 (b) 72
(c) 120 (d) 144
[NDA (I) - 2012]

11. The number of permutations that can be formed from all the letters of the word 'BASEBALL' is:
(a) 540 (b) 1260
(c) 3780 (d) 5040
[NDA (II) - 2012]

12. What is the number of diagonals which can be drawn by joining the angular points of a polygon of 100 sides?
(a) 4850 (b) 4950
(c) 5000 (d) 10000
[NDA-2012(2)]

13. In how many ways can the letters of the word 'GLOOMY' be arranged so that the two O's should not be together?
(a) 240 (b) 480
(c) 60 (d) 720
[NDA (I) - 2013]

14. If P (77, 31) =x and C (77, 31) = y, then which one of the following is correct?
(a) $x = y$ (b) $2x = y$
(c) $77x = 31y$ (d) $x > y$
[NDA (I) - 2013]

15. A bag contains balls of two colours, 3 black and 3 white. What is the smallest number of balls which must be drawn from the bag, without looking, so that among these there are two of the same colour?
(a) 2 (b) 3
(c) 4 (d) 5
[NDA-2013(1)]

16. What is $\sum_{r=0}^{n} C(n,r)$ equal to?
(a) 2^n-1 (b) n
(c) n! (d) 2^n
[NDA-2013(2)]

17. If $C(28,2r) = C(28, 2r-4)$, then what is r equal to?
(a) 7 (b) 8
(c) 12 (d) 16
[NDA-2013(2)]

18. How many different words can be formed by taking four letters out of the letters of the word 'AGAIN' if each word has to start with A ?
(a) 6 (b) 12
(c) 24 (d) None
[NDA (I) - 2014]

19. Out of 7 consonants and 4 vowels, words are to be formed by involving 3 consonants and 2 vowels. The number of such words formed is :
(a) 25200 (b) 22500
(c) 10080 (d) 5040
[NDA (I) - 2014]

For the next three (03) items that follow:
Given that $C(n,r) : C(n, r+1) = 1:2$ and $C(n, r+1) : C(n, r+2) = 2:3$.

20. What is n equal to?
(a) 11 (b) 12
(c) 13 (d) 14
[NDA-2014(1)]

21. What is r equal to?
(a) 2 (b) 3
(c) 4 (d) 5
[NDA-2014(1)]

22. What is $P(n,r) : C(n,r)$ equal to?
(a) 6 (b) 24

(c) 120 (d) 720

[NDA-2014(1)]

23. What is the number of ways in which one can post 5 letters in 7 letters boxes?
(a) 7^5 (b) 3^5
(c) 5^7 (d) 2520

[NDA (II) - 2014]

24. What is the number of ways that a cricket team of 11 players can be made out of 15 players?
(a) 364 (b) 1001
(c) 1365 (d) 32760

[NDA (II) - 2014]

25. How many words can be formed using all the letters of the word 'NATION' so that all the three vowels should never come together?
(a) 354 (b) 348
(c) 288 (d) None of these

[NDA (I) - 2015]

26. What is $\sum_{r=0}^{1} {}^{n+r}C_n$ equal to?
(a) ${}^{n+1}C_1$ (b) ${}^{n+2}C_n$
(c) ${}^{n+3}C_n$ (d) ${}^{n+2}C_{n+1}$

[NDA-2015(1)]

27. The number of ways in which a cricket team of 11 players be chosen out of a batch of 15 players so that the captain of the team is always included, is
(a) 165 (b) 364
(c) 1001 (d) 1365

[NDA (I) - 2015]

28. If different words are formed with all the letters of the word 'AGAIN' and are arranged alphabetically among themselves as in a dictionary, the word at the 50th place will be
(a) NAAGI (b) NAAIG
(c) IAAGN (d) IAANG

[NDA (I) - 2015]

29. The number of 3-digit even numbers that can be formed from the digits 0, 1, 2, 3, 4 and 5, repetition of digits being not allowed, is
(a) 60 (b) 56
(c) 52 (d) 48

[NDA (II) - 2015]

30. The number of ways in which 3 holiday tickets can be given to 20 employees of an organization if each employee is eligible for any one or more of the tickets, is
(a) 1140 (b) 3420
(c) 6840 (d) 8000

NDA (II) - 2015]

31. A polygon has 44 diagonals. The number of its sides is
(a) 11 (b) 10
(c) 8 (d) 7

[NDA (II) - 2015]

32. What is the number of four-digit decimal numbers in which no digit is repeated?
(a) 3024 (b) 4536
(c) 5040 (d) None

[NDA (I) - 2016]

33. What is the number of ways in which 3 holiday travel tickets are to be given to 10 employees of an organization, if each employee is eligible for any one or more of the tickets?
(a) 60 (b) 120
(c) 500 (d) 1000

[NDA (I) - 2016]

34. What is the number of different messages that can be represented by three 0's and two 1's?
(a) 10 (b) 9
(c) 8 (d) 7

[NDA (I) - 2016]

35. What is the number of odd integers between 1000 and 9999 with no digit repeated?
(a) 2100 (b) 2120
(c) 2240 (d) 3331

[NDA (II) - 2016]

36. A five-digit number divisible by 3 is to be formed using the digits 0, 1, 2, 3 and 4 without repetition of digits. What is the number of ways this can be done?
(a) 96
(b) 48
(c) 32
(d) No number can be formed

[NDA (II) - 2016]

37. Out of 15 points in a plane, n points are in the same straight line. 445 triangles can be formed by joining these points. What is the value of n?
(a) 3 (b) 4
(c) 5 (d) 6

[NDA (II) - 2016]

38. What is ${}^{47}C_4 + {}^{51}C_3 + \sum_{j=2}^{5} {}^{52-j}C_3$ equal to?
(a) ${}^{52}C_4$ (b) ${}^{51}C_5$
(c) ${}^{53}C_4$ (d) ${}^{52}C_5$

[NDA (II) - 2016]

39. The number of different words (eight-letter words) ending and beginning with a consonant which can be made out of the letters of the word 'EQUATION' is
(a) 5200 (b) 4320
(c) 3000 (d) 2160

[NDA (I) - 2017]

40. Let A = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}. Then the number of subsets of A containing two or three elements is:
(a) 45 (b) 120
(c) 165 (d) 330

[NDA-2017(1)]

41. Three-digit numbers are formed from the digits 1, 2 and 3 in such a way that the digits are not repeated. What is the sum of such three-digit numbers?
(a) 1233 (b) 1322
(c) 1323 (d) 1332

[NDA-2017(1)]

42. The value of [C(7,0) + C(7,1)] + [C(7,1) + C(7,2)] +……………..+[C(7,6)+C(7,7)] is
(a) 254 (b) 255
(c) 256 (d) 257

[NDA-2017(1)]

43. A tea party is arranged for 16 people along two sides of a long table with eight chairs on each side. Four particular men wish to sit on one particular side and two particular men on the other side. The number of ways they can be seated is
(a) $24 \times 8! \times 8!$ (b) $(81)^3$
(c) $210 \times 8! \times 8!$ (d) 16!

[NDA (II) - 2017]

44. How many different permutations can be made out of the letters of the word 'PERMUTATION'?
(a) 19958400 (b) 19954800
(c) 19952400 (d) 39916800

[NDA (II) - 2017]

45. What is the number of triangles that can be formed by choosing the vertices from a set of 12 points in a plane seven of which lie on the same straight line?
(a) 185 (b) 175
(c) 115 (d) 105
[NDA (I) - 2018]

46. How many four-digit numbers divisible by 10 can be formed using 1, 5, 0, 6, 7 without repetition of digits?
(a) 24 (b) 36
(c) 44 (d) 64
[NDA (I) - 2018]

47. How many numbers between 100 an 1000 can be formed with the digits 5, 6, 7, 8, 9, if the repetition of digits is not allowed?
(a) 32 (b) 5^3
(c) 120 (d) 60
[NDA (I) - 2018]

48. What is C(n, r) + 2C (n, r–1) + C(n, r–2) equal to?
(a) C(n+1, r) (b) C(n–1, r+1)
(c) C(n, r+1) (d) C(n+2, r)
[NDA-2018(1)]

49. Let x be the number of integers lying between 2999 and 8001 which have at least two digits equal. Then x is equal to:
(a) 2480 (b) 2481
(c) 2482 (d) 2483
[NDA-2018(2)]

50. What is the sum of all three-digit numbers that can be formed using all the digits 3, 4 and 5, when repetition of digits is not allowed?
(a) 2664 (b) 3882
(c) 4044 (d) 4444
[NDA (II) - 2018]

51. The total number of 5-digit numbers that can be composed of distinct digits from 0 to 9 is
(a) 45360 (b) 30240
(c) 27216 (d) 15120
[NDA (II) - 2018]

52. There are 17 cricket players, out of which 5 players can bowl .In how many ways can a team of 11 players be selected to as to include 3 bowlers?
(a) C(17,11) (b) C(12,8)
(c) C(17,5)×C(5,3) (d) C(5,3) ×C(12,8)
[NDA (II) - 2018]

53. How many three-digit even numbers can be formed using the digits 1, 2, 3, 4 and 5 when repetition of digits is not allowed?
(a) 36 (b) 30
(c) 24 (d) 12
[NDA (I) - 2019]

54. From 6 programmers and 4 typists, an office wants to recruit 5 people. What is the number of ways this can be done so as to recruit at least one typist?
(a) 209 (b) 210
(c) 246 (d) 242
[NDA (I) - 2019]

55. There are 10 points in a plane. No three of these points are in a straight line. What is the total number of straight lines which can be formed by joining the points?
(a) 90 (b) 45
(c) 40 (d) 30
[NDA (I) - 2019]

56. If c(20, n+2) = C(20, n–2), then what is n equal to?
(a) 8 (b) 10
(c) 12 (d) 16
[NDA-2019(1)]

57. What is C(47,4) + C(51,3) + C(50,3) + C(49,3) + C(48,3) + C(47,3) equal to?
(a) C(47,4) (b) C(52,5)
(c) C(52,4) (d) C(47,5)
[NDA – 2019(2)]

58. If n! has 17 zeros, then what is the value of n?
(a) 95
(b) 85
(c) 80
(d) no such value of n exists
[NDA-2019(2)]

59. If P(n,r) = 2520 and C(n,r) = 21, then what is the value of C(n + 1, r + 1) ?
(a) 7 (b) 14
(c) 28 (d) 56
[NDA (II) - 2019]

60. What is the number of diagonals of an octagon?
(a) 48 (b) 40
(c) 28 (d) 20
[NDA-2019(2)]

61. If C(20, n + 2) = C(20, n − 2) , then what is n equal to ?
(a) 18 (b) 25
(c) 10 (d) 11
[NDA 2020]

62. What is the number of ways in which the letters of the word 'ABLE' can be arranged so that the vowels occupy odd places?
(a) 2 (b) 4
(c) 6 (d) 8
[NDA 2020]

63. What is the maximum number of points of intersection of 5 non-overlapping circles
(a) 10 (b) 15
(c) 20 (d) 25
[NDA 2020]

64. How many 5-digit prime numbers can be formed by using the digits 1, 2, 3, 4 5. If the repetition of digits is not allowed?
(a) 5 (b) 4
(c) 3 (d) 0
[NDA (I) - 2021]

65. In how many ways can a team of 5 players be selected from 8 players so as not to include a particular player?
(a) 42 (b) 35
(c) 21 (d) 20
[NDA (I) - 2021]

66. If $a_n = n\,(n!)$, then what is $a_1 + a_2 + a_3 + \ldots + a_{10}$
(a) 10! – 1 (b) 11! + 1
(c) 10! + 1 (d) 11! – 1
[NDA (II) 2021]

67. Let S = (2, 3, 4, 5, 6, 7, 9). How many different 3-digit numbers (with all digits different) from S can be made which are less than 500?
(a) 30 (b) 49
(c) 90 (d) 147
[NDA (II) - 2021]

68. Consider the digit 3, 5, 7, 9. What is the number of 5-digit numbers formed by these digits in which each of these four digits appears?
(a) 240 (b) 180
(c) 120 (d) 60
[NDA (II) - 2021]

69. If C (n, 4),C(n, 5) and C(n, 6) are in AP, then what is the value of n?

(a) 7 (b) 8
(c) 9 (d) 10

[NDA (II) - 2021]

70. How many 4 letter words (with or without meaning) containing two vowels can be constructed using only the letters (without repetition) of the word 'LUCKNOW'?
(a) 240 (b) 200
(c) 150 (d) 120

[NDA (II) - 2021]

71. Suppose 20 distinct points are placed randomly on a circle. Which of the following statements is/are correct?
1.The number of straight lines that can be drawn by joining any two of these points is 380.
2.The number of triangles that can be drawn by joining any three of these points is 1140.
Select the correct answer using the code given below:
(a)1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (II) - 2021]

72. Consider a regular polygon with 10 sides. What is the number of triangles that can be formed by joining the vertices which have no common side with any of the sides of the polygon?
(a) 25 (b) 50
(c) 75 (d) 100

[NDA (II) - 2021]

73. If C(3n, 2n) = C(3n, 2n–7), then what is the value of C(n, n–5)?
(a) 42 (b) 35
(c) 28 (d) 21

[NDA (I) - 2022]

74. What is the value of
C(51,21)–C(51,22)+C(51,23)–C(51,24) +C(51,25)–C(51,26)+C(51,27)–C(51,28) +C(51,29)–C(51,30)?
(a) C(51,25) (b) C(51, 27)
(c) C(51,51) – C(51,0) (d) C(51,25) – C(51,27)

[NDA (I) - 2022]

75. How many odd numbers between 300 and 400 are there in which none of the digits is repeated?
(a) 32 (b) 36
(c) 40 (d) 45

[NDA (I) - 2022]

76. Consider the following statements:
1. $\frac{n!}{3!}$ is divisible by 6, where n > 3
2. $\frac{n!}{3!}+3$ is divisible by 7, where n > 3
Which of the above statements is/are correct?
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (I) - 2022]

77. In how many ways can a term of 5 players be selected out of 9 players so as to exclude two particular players?
(a) 14 (b) 21
(c) 35 (d) 42

[NDA (I) - 2022]

78. How many permutations are there of the letters of the word 'TIGER' in which the vowels should not occupy the even positions?
(a) 72 (b) 36
(c) 18 (d) 12

[NDA (I) - 2022]

79. What is the maximum value of n such that 5^n divides (30!+35!), where n is a natural number?
(a) 4 (b) 6
(c) 7 (d) 8

[NDA (I) - 2022]

80. What is the value of 2 (2×1) + 3 (3 × 2 × 1) + 4 (4 × 3 × 2 × 1) + 5 (5 × 4 × 3 × 2 × 1) + …. …. …. + 9 (9 × 8 × 7 × 6 × 5 × 4 × 3 × 2 × 1) + 2?
(a) 11! (b) 10!
(c) 10 + 10! (d) 11 + 10!

[NDA (I) - 2022]

81. How many four digit natural numbers are there such that all of the digits are odd?
(a) 625 (b) 400
(c) 196 (d) 120

[NDA 2022 (II)]

82. If different permutations of the letters of the word 'MATHEMATICS' are listed as in a dictionary, how many words (with or without meaning) are there in the list before the first word that starts with C?
(a) 302400 (b) 403600
(c) 907200 (d) 1814400

[NDA 2022 (II)]

Consider the following for the next three (03) items that follow:
Consider the word 'QUESTION':

83. How many 4-letter words each of two vowels and two consonants with or without meaning, can be formed?
(a) 36 (b) 144
(c) 576 (d) 864

[NDA 2022 (II)]

84. How many 8-letter words with or without meaning, can be formed such that consonants and vowels occupy alternate positions?
(a) 288 (b) 576
(c) 1152 (d) 2304

[NDA 2022 (II)]

85. How many 8-letter words with or without meaning, can be formed so that all consonants are together?
(a) 5760 (b) 2880
(c) 1440 (d) 720

[NDA 2022 (II)]

86. Consider the following statements for a fixed natural number n:
1.C(n, r) is greatest if n = 2r
2.C(n, r) is greatest if n = 2r – 1 and n = 2r + 1.
Which of the statements given above is/are correct?
(a) 1 only (b) 2 only
(c) both 1 and 2 (d) Neither 1 nor 2

[NDA – 2023 (1)]

87. M parallel lines cut n parallel lines giving rise to 60 parallelograms. What is the value of (m + n)?
(a) 6 (b) 7
(c) 8 (d) 9

[NDA – 2023 (1)]

88. Let x be the number of permutation of the word 'PERMUTATIONS' and y be the number of permutations of the word 'COMBINATIONS'. Which one of the following is correct?
(a) x = y (b) y = 2x
(c) x = 4y (d) y = 4x

[NDA – 2023 (1)]

89. 5-digit numbers are formed using the digits 0, 1, 2, 4, 5 without repetition. What is the percentage of numbers which are greater than 50,000.
(a) 20% (b) 25%

(c) $\frac{100}{3}\%$ (d) $\frac{110}{3}\%$

[NDA – 2023 (1)]

90. Consider the following statements:
1.(25)! + 1 is divisible by 26
2.(6)! + 1 is divisible by 7
Which of the above statement is/are correct?
(a) 1 only (b) 2 only
(c) both 1 and 2 (d) neither 1 nor 2

[NDA – 2023 (1)]

91. What is the number of 6-digit numbers that can be formed only by using 0, 1, 2, 3, 4 and 5 (each once); and divisible by 6?
(a) 96 (b) 120
(c) 192 (d) 312

[NDA – 2023 (1)]

Consider the following for the next two (02) items that follow:
Consider the sum S = 0! + 1! + 2! + 3! + 4! + …… + 100!

92. If the sum S is divided by 8, what is the remainder?
(a) 0
(b) 1
(c) 2
(d) cannot be determined

[NDA – 2023 (1)]

93. If the sum S is divided by 60, what is the remainder?
(a) 1 (b) 3
(c) 17 (d) 34

[NDA – 2023 (1)]

94. If 1! + 3! + 5! + 7! + …..+ 199! is divided by 24, what is the remainder?
(a) 3 (b) 6
(c) 7 (d) 9

[NDA-2023 (2)]

95. What is the maximum number of points of intersection of 10 circles?
(a) 45 (b) 60
(c) 90 (d) 120

[NDA-2023 (2)]

96. How many four-digit natural numbers are there such that all of the digits are even?
(a) 625 (b) 500
(c) 400 (d) 256

[NDA-2024 (1)]

97. Four digit numbers are formed by using the digits 1, 2, 3, 5 without repetition of digits. How many of them are divisible by 4?
(a) 120 (b) 24
(c) 12 (d) 6

[NDA-2024 (1)]

98. What is the sum of all four digit numbers formed by using all digits 0, 1, 4, 5 without repetition of digits?
(a) 44440 (b) 46460
(c) 46440 (d) 64440

[NDA-2024 (1)]

99. A man has 7 relatives (4 women and 3 men). His wife also has 7 relatives (3 women and 4 men). In how many wasy can they invite 3 women and 3 men so that 3 of them are men's relatives and 3 of them are his wife's relatives?
(a) 340 (b) 484
(c) 485 (d) 469

[NDA-2024 (1)]

100. A triangle PQR is such that 3 points lie on the side PQ, 4 points on QR and 5 points on RP respectively. Triangles are constructed using these points as vertices. What is the number of triangles so formed?
(a) 205 (b) 206
(c) 215 (d) 220

[NDA-2024 (1)]

101. If 26! = $n8^k$, where k and n are positive integers, then what is the maximum value of k?
(a) 6 (b) 7
(c) 8 (d) 9

[NDA-2024 (1)]

102. What is the maximum number of possible points of intersection of four straight lines and a circle (intersection is between lines as well as circle and lines) ?
(a) 6 (b) 10
(c) 14 (d) 16

[NDA-2024 (2)]

103. In how many ways can the letters of the word INDIA be permutated such that in each combination, vowels should occupy odd position?
(a) 3 (b) 6
(c) 9 (d) 12

[NDA-2024 (2)]

104. The letter of the word EQUATION are arranged in such a way that all vowels as well as consonants are together. How many such arrangements are there?
(a) 240 (b) 720
(c) 1440 (d) 1620

[NDA-2024 (2)]

105. In how many ways can a student choose (n – 2) courses out of n courses if 2 courses are compulsory?
(a) (n – 3) (n – 4) (b) (n – 1) (n – 2)
(c) (n – 3) (n – 4)/2 (d) (n – 2) (n – 3)/2

[NDA-2024 (2)]

106. How many four digit numbers are there having all digits are odd?
(a) 625 (b) 400
(c) 196 (d) 120

[NDA-2024 (2)]

107. How many 7-letter words (with or without meaning) can be constructed using all the letters of the word CAPITAL so that all consonants come together in each word?
(a) 360 (b) 300
(c) 288 (d) 240

[NDA-2025 (1)]

108. How many sides are there in a polygon which has 20 diagonals?
(a) 6 (b) 7
(c) 8 (d) 10

[NDA-2025 (1)]

109. In how many ways can the letters of the word DELHI be arranged keeping the positions of vowels and consonants unchanged?
(a) 6 (b) 9
(c) 12 (d) 24

[NDA-2025 (1)]

110. What is the number of positive integer solutions of $x + y + z = 5$?
(a) 3 (b) 5
(c) 6 (d) 9

[NDA-2025 (1)]

111. If the number of selections of r as well as $(n + r)$ things from $5n$ different things are equal, then what is the value of r?
(a) n (b) 2n
(c) 3n (d) 4n

[NDA-2025 (1)]

112. What is the number of selections of at most 3 things from 6 different things?
(a) 20 (b) 22
(c) 41 (d) 42
[NDA-2025 (1)]

113. Let y = x! and z = (2x)!. If (z/y) = 120, then what is the value of (3x)!?
(a) 362880 (b) 181440
(c) 90720 (d) 45360
[NDA-2025 (2)]

114. Let n be a natural number. The number of consecutive zeros at the end of the expansion of n! is exactly 2. How many values of n are possible?
(a) 3 (b) 4
(c) 5 (d) More than 5
[NDA-2025 (2)]

115. How many 4-digit numbers that are divisible by 4 can be formed using the digits 1, 2, 3 and 4 (repetition of digits is not allowed)?
(a) 3 (b) 6
(c) 9 (d) 12
[NDA-2025 (2)]

For the following two (02) items:
There are 8 points on a plane out of which 4 points are collinear.

116. How many triangles can be formed by joining these points?
(a) 56 (b) 54
(c) 53 (d) 52
[NDA-2025 (2)]

117. How many quadrilaterals can be formed by joining these points?
(a) 70 (b) 69
(c) 53 (d) None of the above
[NDA-2025 (2)]

ANSWER KEY

1.	b	**2.**	b	**3.**	c	**4.**	d	**5.**	b	**6.**	d	**7.**	a	**8.**	d	**9.**	b	**10.**	d
11.	d	**12.**	a	**13.**	a	**14.**	d	**15.**	a	**16.**	d	**17.**	b	**18.**	c	**19.**	a	**20.**	d
21.	c	**22**	b	**23.**	a	**24.**	c	**25.**	c	**26.**	d	**27.**	c	**28.**	b	**29.**	c	**30.**	d
31.	a	**32.**	b	**33.**	d	**34.**	a	**35.**	c	**36.**	d	**37.**	c	**38.**	a	**39.**	b	**40.**	c
41.	d	**42.**	a	**43.**	c	**44.**	a	**45.**	a	**46.**	a	**47.**	d	**48.**	d	**49.**	b	**50.**	a
51.	c	**52.**	d	**53.**	c	**54.**	c	**55.**	b	**56.**	b	**57.**	c	**58.**	d	**59.**	c	**60.**	d
61.	c	**62.**	b	**63.**	c	**64.**	d	**65.**	c	**66.**	d	**67.**	c	**68.**	a	**69.**	a	**70.**	a
71.	b	**72.**	b	**73.**	d	**74.**	c	**75.**	a	**76.**	d	**77.**	b	**78.**	b	**79.**	c	**80.**	b
81.	a	**82.**	c	**83.**	d	**84.**	c	**85.**	b	**86.**	c	**87.**	d	**88.**	c	**89.**	b	**90.**	b
91.	d	**92.**	c	**93.**	d	**94.**	c	**95.**	c	**96.**	b	**97.**	d	**98.**	d	**99.**	c	**100.**	a
101.	b	**102.**	c	**103.**	b	**104.**	c	**105.**	d	**106.**	a	**107.**	c	**108.**	c	**109.**	c	**110.**	c
111.	b	**112.**	d	**113.**	a	**114.**	c	**115.**	b	**116.**	d	**117.**	c						

Solutions

Sol.1. (b)
Number of triangles using 5 points out of three are on a
Straight line = ${}^5C_3 - {}^5C_3$ =
$\frac{5!}{3!2!} - 1$
= 10 − 1 = 9

Sol.2. (b)
We have to construct 5 digit numbers which are greater than 41000. So, we have only 2 ways to choose 5th digit.

5th	4th	3rd	2nd	1st

[∴ Only 4 or 5 can come at 5th place]
Thus, for 4th place we have 4 ways to choose digits,
For 3rd place we have 3 ways.
For 2nd place we have 2 ways
And for unit place we have only 1 way.
Required number of ways=2 x 4 x 3 x 2 x 1=48

Sol.3. (c)
Let P (15,n-1) : P(16, n-2) = 3:4

$$\Rightarrow \frac{{}^{15}P_{n-1}}{{}^{16}P_{n-2}} = \frac{3}{4}$$

$$\Rightarrow \frac{15!}{(15-n+1)!} \times \frac{(16-n+2)}{16!} = \frac{3}{4}$$

$$\Rightarrow \frac{15!}{(16-n)!} \times \frac{(18-n)!}{16!} = \frac{3}{4}$$

$$\Rightarrow \frac{(18-n)!}{16(16-n)!} \times \frac{3}{4}$$

$$\Rightarrow \frac{(18-n)(17-n)(16-n)!}{16(16-n)!} = \frac{3}{4}$$

$$\Rightarrow (18-n)(17-n=12)$$

$$\Rightarrow 306-17n-18n+n^2=12$$

$$\Rightarrow n^2-35n+294=0$$

$$\Rightarrow (n-14)(n-21)=0$$

$\Rightarrow n=14$ (∵

Sol.4. (d)
Required number of ways= ${}^6C_2 \times 2$
= 6 × 5 = 30

Sol.5. (b)
Number of ways when one specified book is always included. So, 4 books can be chose from remaining 9 ways:
= m = ${}^9C_4$
Number of ways when one specified book is always excluded. So, 5 books can be chose from remaining 9 ways:
= n = ${}^9C_5$
$\Rightarrow m = n$

Sol.6. (d)
Since, combinations of taking 1,2,3,….things at a times are ${}^nC_1$, ${}^nC_2$, ${}^nC_3$ .... ${}^nC_n$
∴ Total number of combinations
$= {}^nC_1 + {}^nC_2 + ... + {}^nC_n = 2^n - 1$

Sol.7. (a)
Consider $\sum_{r=1}^{n} \frac{P(n,r)}{r!} = \sum_{r=1}^{n} C(n,r)$
$= {}^nC_1 + {}^nC_2 + ... + {}^nC_n = 2^n - 1$

Sol.8. (d)
There are 4 candidates.
This means there are 4 blanks spaces for 1 post.
Now, that 1 post is to be selected by votes of 5 men.
So, All 4 places can be filling by each man's votes.
∴ There is 5 ways for 1 place.
=5 x 5 x 5 x 5 = 625. (∵ we have 4 places)

Sol.9. (b)
Number of diagonals = $\frac{n(n-3)}{2}$

Sol.10. (d)
BGBGBGB
Required no. of ways = 4! x 3! = 144

Sol.11. (d)
There are total 8 letters in the word BASEBALL, in which we have 2 B's 2A's and 2 L's.
∴ Required No. of permutations= $\frac{8!}{2!\times 2!\times 2!}$
$= \frac{8\times7\times6\times5\times4\times3\times2\times1}{8} = 5040$

Sol.12. (a)
no.of diagonals
$= \frac{n(n-3)}{2} = \frac{100\times97}{2} = 4850$

Sol.13. (a)
total words formed by GLOOMY – words formed by taking both O together
$\frac{6!}{2!} - 5! = 240$

Sol.14. (d)
As we know P(n,r)=r!C(n,r)
∴ From the question, we have
x = r!(y)
Here r = 31
x = (31)! y
so x > y

Sol.15. (b)
if we take 2 balls together then it may be of different colour
if we take 3 balls together then minimum 2 balls must be of same colour.

Sol.16. (d)
$\sum_{r=0}^{n} C(n,r) = {}^nC_0 + {}^nC_1 + {}^nC_2 + ... + {}^nC_n$
= 2^n

Sol.17. (b)
${}^{28}C_{2r} = {}^{28}C_{2r-4}$
2r + 2r – 4 = 28
4r = 32
r = 8

Sol.18. (c)
As 'A' must be first letter of each word.
Total number of words = 4! = 24

Sol.19. (a)
Number of words = ${}^7C_3$ x ${}^4C_2$x 5!
$= 120 \times \frac{7!}{4!3!} \times \frac{4!}{2!2!} = 25200$

Sol.20. (d)

$$\frac{{}^nC_r}{{}^nC_{r+1}} = \frac{1}{2} \Rightarrow \frac{\frac{n!}{r!(n-r)!}}{\frac{n!}{(r+1)!(n-r-1)!}} = \frac{1}{2}$$

$$\Rightarrow \frac{r+1}{n-r} = \frac{1}{2}$$

$\Rightarrow n - 3r = 2$ ……….(i)

$$\frac{{}^nC_{r+1}}{{}^nC_{r+2}} = \frac{2}{3} \Rightarrow \frac{\frac{n!}{(r+1)!(n-r-1)!}}{\frac{n!}{(r+2)!(n-r-2)!}} = \frac{2}{3}$$

$$\Rightarrow \frac{r+2}{n-r-1} = \frac{2}{3}$$

$\Rightarrow 2n - 5r = 8$ ………(ii)
by eq (i) and (ii)
n = 14 and r = 4

Sol.21. (c)
by above solution r = 4

Sol.22. (b)
$\frac{{}^nP_r}{{}^nC_r} = r! = 4! = 24$

Sol.23. (a)
First letter can be put any 7 letters boxes=7 ways
Similarly, 2nd, 3rd, 4th and 5th letter be put in 7 ways each respectively
∴7 x 7 x 7 x 7 x 7=7^5

Sol.24. (c)
Number of ways that a cricket team of 11 players can be made out of 15 players
${}^{15}C_{11} = \frac{15!}{11!4!}$
$= \frac{15\times14\times13\times12\times11!}{11!\times1\times2\times3\times4} = 1365$

Sol.25. (c)
The given word is 'NATION'
Total number of words that can be formed form given word 'NATION'

$$= \frac{6!}{2!} = \frac{6\times5\times4\times3\times2!}{2!} = 360$$

Now numbers of word that can be formed from given word NATION, so that all vowels never comes together.
$= 360 - \left[4!\frac{3!}{2!}\right] = 360 - [24\times3]$
$= 360 - 72 = 288$
∴Option (c) is correct

Sol.26. (d)
$\sum_{r=0}^{1} {}^{n+r}C_n = {}^nC_n + {}^{n+1}C_n = 1 + n + 1 = n + 2$
from options
${}^{n+2}C_{n+1} = n + 2$

Sol.27. (c)
If captain is always included then we can choose 10 more players out of the remaining 14 players.
So
${}^{14}P_{10} = \frac{14!}{10!4!} = 1001$

Sol.28. (b)
First two words No. of words

AA	3!=6
AG	3!=6
AI	3!=6
AN	3!=6
GA	3!=6
GI	3!/2!=3
GN	3!/2!=3
IA	3!=6
IG	3!/2!=3
IN	3!/2!=3
NA	3!=6
	Total = 54

It means 50th word will be starting with 'NA'

N A A G I (49th Place)
N A A I G (50th Place)
N A G A I (51st Place)
N A G I A (52nd Place)
N A I A G (53rd Place)
N A I G A (54th Place)

Sol.29. (c)
No. of digits to be filled at one's place=3
No. of digits to be filled at 10's place=5 No.of digits to be filled at 100's place=4
$\therefore$ No. of digits formed with zero at 100's place = 1 x 2 x 4 = 8
$\therefore$ Required no. of digits formed = 60 - 8 = 52

Sol.30. (d)
Each employee is eligible for 1 or more of the tickets.
$\therefore$ No. of ways =20 x 20 x 20 = 8000.

Sol.31. (a)
No. of diagonals in a polygon

$$\Rightarrow 44=\frac{n(n-3)}{2}$$
$$\Rightarrow n^2-3n-88=0$$
$$\Rightarrow (n-11)(n+8)=0$$

n = 11

Sol.32. (b)
Let the given 4 digit decimal number is
Places after decimal can be filled in the following ways

7 8 9 9

Total number of ways = 7 x 8 x 9 x 9 = 4536

Sol.33. (d)
No. of ways in which 3 holidays travel tickets are to be given to 10
Employees = 10^3 = 1000

Sol.34. (a)
Number of different messages that can be represented by three 0's and two 1's is

$$\frac{5!}{3!\text{x}2!}=10$$

Option (a) is correct.

Sol.35. (c)

8	8	7	5

at 4th place there are 5 possibilities 1,3,5,7,9
at 1st 0 and digit and 4th place can not repeat so no. of possibilities = 8
at 2nd no. at 1st and 4th cannot repeat so no. of possibilities = 8
at 3rd no. at 1st , 2nd and 4th cannot repeat so no. of possibilities = 7
Total No's = 8 x 8 x 7 x 5 = 2240

Sol.36. (d)
Since sum of digits= 10 (which is not divisible by)
$\therefore$ No numbers can be formed

Sol.37. (c)
Here, ${}^{15}C_3$-${}^{n}C_3$ = 445
455 - ${}^{n}C_3$ = 445
${}^{n}C_3 = 10$
n = 5

Sol.38. (a)
${}^{47}C_3+{}^{47}C_4+{}^{48}C_3+{}^{49}C_3+{}^{50}C_3+{}^{51}C_3+{}^{52}C_3$
${}^{48}C_4+{}^{48}C_3+{}^{49}C_3+{}^{50}C_3+{}^{51}C_3+{}^{52}C_3$
${}^{49}C_4+{}^{49}C_3+{}^{50}C_3+{}^{51}C_3+{}^{52}C_3$
${}^{50}C_4+{}^{50}C_3+{}^{51}C_3+{}^{52}C_3$
${}^{51}C_4+{}^{51}C_3$
${}^{52}C_4$

Sol.39. (b)
EQUATION -8 letters.
Consonants – Q,T, N-3 letters.
First letter of 8-letter word can be any of 3 consonants
Last letter of 8-letter word can be remaining 2 consonants.
The middle 6-letters can be remaining 2 Consonants.
The middle 6-letters can be arranged in 6! ways.
So, number of different words = 3×2×6!
= 6×720 = 4320

Sol.40. (c)
${}^{10}C_2+{}^{10}C_3$ = 45 + 120 = 165

Sol.41. (d)
There will be 6 numbers
Whose sum
123+132+213+231+312+321=1332

Sol.42. (a)
$C_0+2[{}^{7}C_1+{}^{7}C_2+{}^{7}C_3+{}^{7}C_4+\ldots\ldots {}^{7}C_6]+{}^{7}C_7$
$2.2^7-2=2^8-2=256-2=254$

Sol.43. (c)
Number of ways =

$$=\frac{8!8!}{4!6!}\times 10!$$
$$=(8!)^2\times\frac{10!}{4!6!}$$
$$=(8!)^2\times\frac{10\times9\times8\times7}{4!}$$
$$=(8!)^2\times\frac{10\times9\times8\times7}{4\times3\times2\times1}$$

Sol.44. (a)
PERMUTATION
11 letters and t is repeated 2 time.
$\therefore$ different permutations $=\frac{11!}{2!}$
= 11 × 10 × 9 × 8 × 7 × 6 × 5 × 4 × 3
=19958400

Sol.45. (a)
To form a triangle, we need 3 points, 12 points are given So, ${}^{12}P_3$ triangles can be formed.
But, given that 7 points are on a straight line. Selecting 3 points from this set will not form a triangle.
So, number of triangles formed ${}^{12}C_3$–${}^{7}C_3$

$$=\frac{12!}{3!9!}-\frac{7!}{3!4!}$$
$$=\frac{12\times11\times10}{3\times2\times1}-\frac{7\times6\times5}{3\times2\times1}=220-35=185$$

Sol.46. (a)
(a) A number divisible by 10 means the last digit is 0.
So, the remaining 3 digits can be arranged in 4×3×2 ways = 24 ways

Sol.47. (d)
Number between 100 and 1000 are 3-digit numbers. It is given that the digits should not be repeated.
Number of given digits = 5
In a 3-digit number, first number can be arranged in 5 ways.
$\therefore$ Numbers that can be formed =5 × 4 × 3 = 60

Sol.48. (d)
$({}^{n}C_r+{}^{n}C_{r-1})+({}^{n}C_{r-1}+{}^{n}C_{r-2})$
${}^{n+1}C_r+{}^{n+1}C_{r-1}$
${}^{n+2}C_r$

Sol.49. (b)
Total numbers from between 2999 and 8001 are 5001
Numbers with no repetition = 5x9x8x7 = 2520
Numbers with at least 2 digits repeated = 5001-2520 = 2481

Sol.50. (a)
3 digit number made from digit 3, 4 and 5 and having all distinct digit=3!=6 and sum of such numbers are 543 + 534 + 345 + 354 + 435 + 453 = 2664

Sol.51. (c)
Number of 5 digits numbers with all distinct digit is same as filling of 5 vacant placed out of 10 boxes. First digit of any number can be chosen in 9 ways. Remaining 4 digits can be chosen in remaining 9 digits in ${}^{9}P_4$ ways.
Total number of such number
= 9 × 9 × 8 × 7 × 6 = 27216

Sol.52. (d)
3 bowlers are selected among 5 bowlers in ${}^{5}C_3$ ways.
Remaining 8 player's are selected from 12 player's in ${}^{12}C_8$ ways.
$\therefore$ total number of ways = ${}^{12}C_8\times{}^{5}C_3$

Sol.53. (d)
Given digits are 1,2,3,4 and 5
Total number of 3-digit even numbers
= 4 x 3 x 2 = 24

Sol.54. (c)
Number of ways ${}^{4}C_1{}^{6}C_4+{}^{4}C_2{}^{6}C_3+{}^{4}C_3{}^{6}C_2+{}^{4}C_4{}^{6}C_1$
= (4) (15) + (6) (20) + (4) (15) + (1) (6)
= 60 + 120 + 60 + 6 = 246

Sol.55. (b)
A straight line can be formed by joining 2 points.
$\therefore$ Total number of straight lines=${}^{10}C_2$ = 45

Sol.56. (b)
${}^{20}C_{n+2}={}^{20}C_{n-2}$
n + 2 + n – 2 = 20
n = 10

Sol.57. (c)
${}^{n}C_r+{}^{n}C_{r+1}={}^{n+1}C_{r+1}$
$=({}^{47}C_3+{}^{47}C_4)+{}^{48}C_3+{}^{49}C_3+{}^{50}C_3+{}^{51}C_3$
$=({}^{48}C_4+{}^{48}C_3)+{}^{49}C_3+{}^{50}C_3+{}^{51}C_3$
$=({}^{49}C_4+{}^{49}C_3)+{}^{50}C_3+{}^{51}C_3$
$=({}^{50}C_4+{}^{50}C_3)+{}^{51}C_3$
$=({}^{51}C_3+{}^{51}C_3)={}^{52}C_4$

Sol.58. (d)

5	75
5	15
	3

18 zeros in 75!

5	74
5	14
	2

16 zeros in 74!
17 zeros not possible.

Sol.59. (c)
${}^{n}C_r\times r!={}^{n}P_r$
$\Rightarrow 21\times r!=2520$
$\Rightarrow r!=120\Rightarrow r=5$
${}^{n}P_r=2520$

$\Rightarrow \frac{n!}{(n-5)!} = 2520$

$\Rightarrow n = 7$

$^8C_6 = 28$

Sol.60. (d)

no. of diagonals = $\frac{n(n-3)}{2} = \frac{8\times5}{2} = 20$

Sol.61. (c)

C(20, n + 2) = C(20 , n−2)

$^{20}C_{n-2} = {}^{20}C_{n+2}$

$[{}^nC_r = {}^nC_{n-r}]$

n − 2 + n + 2 = 20

n = 10

Sol.62. (b)

CVCV

Vowels can occupy 2 places and consonants may also occupy 2 places

2!2! = 4

Sol.63. (c)

Each two circles will intersect at 2 places

so number of intersections = $2.\,{}^5C_2 = 20$

Sol.64. (d)

sum of 1,2,3,4,5 is 15.

so number formed by these integers is always divisible by 3.

Sol.65. (c)

one excluded so $^7C_5 = 21$

Sol.66. (d)

0(0!)+ 1(1!)+2(2!)+3(3!)+……..+10(10!)

(1! - 0!)+ (2! - 1!)+ (3! - 2!)+…..+(11! - 10!)

=11! - 1

Sol.67. (c)

↑	↑	↑

3 × 6 × 5 = 90

Sol.68. (d)

3, 5, 7, 9 digits are given to us,

We have to find 5 dight no's from these digit

3579↓

$\therefore \frac{5!}{2!} = \frac{5\times4\times3\times2!}{2!} = 60$

Here one position if left so one of the no. will repeat itself

Sol.69. (a)

$^nC_4$, $^nC_5$, $^nC_6$, are in AP

2. $^nC_5 = {}^nC_4 + {}^nC_6$

$2.\ \frac{n!}{5!(n-5)!} = \frac{n!}{4!(n-4)!} + \frac{n!}{6!(n-6)!}$

$\frac{1}{5.4!(x-5)(n-6)!} = \frac{1}{4(x-4)(x-5)!(n-6)} + \frac{1}{6.5.4!(n-6)!}$

By solving above quadratic equation ⇒ n = 7

Sol.70. (a)

There are 5 consonants and 2 vowel 2 vessel and 2 consonants are selected as $^5C_2 \times {}^2C_2$ = 10 types.

These 4 letters can arrange as 4! =24

Total possible cases = 10 × 24 = 240

Sol.71. (b)

Number of straight lines = $^{20}C_2 = \frac{20\times19}{2} = 190$

Number of triangles = $^{20}C_3 = \frac{20\times19\times18}{3\times2\times1} = 1140$

Sol.72. (d)

Total possible triangle = $^{10}C_3 = 120$

Triangle with 1 common side of polygon = 60

Triangle with 2 common sides of polygon = 10

Triangle with no common side = 120 – 60 – 10 = 50

Sol.73. (d)

$^{3n}C_{2n} = {}^{3n}C_{2n-7}$

2n + 2n–7 = 3n

n=7

$^nC_{n-5} = {}^7C_2 = 21$

Sol.74. (c)

$^nC_r = {}^nC_{n-r}$

$^{51}C_{20} = {}^{51}C_{31}$, $^{51}C_{21} = {}^{51}C_{30}$ and so on.

Sum will be zero

Sol.75. (a)

3		1 5 7 9
↑	↑	↑

1 × 8 × 4 = 32

Sol.76. (d)

Statement 1 is not correct for n=4

Statement 2 is not correct for n=5

Sol.77. (b)

5 player out of 7

$= {}^7C_5 = 21$

Sol.78. (b)

O	E	O	E	O

$\Rightarrow {}^3C_2 \times 2! \times 3!$

⇒3 × 2 × 6 = 36

Sol.79. (c)

30! + 35!

=30! (1+35 × 34 × 33 × 32 × 31)

30! Contains

7 times factor 5

Sol.80. (b)

2(2 × 1) = 3!– 2!

3(3 × 2 × 1) 4! –3!

4(4 × 3 × 2 ×1) = 5!– 4!

Ans so on

(3!– 2!)+(4! –3!)+(5!– 4!)+…(10!– 9!)+2

= 10!

Sol.81. (a)

we have digits 1, 3, 5, 7, 9

number of cases = 5 × 5 × 5 × 5 = 625

Sol.82. (c)

for words starting from A , fix A at starting so

number of words $\Rightarrow \frac{10!}{2!.2!.} = 907200$

Sol.83. (d)

choose any 2 consonants and 2 vowels then arrange these 4 alphabets

$^4C_2 \times {}^4C_2 \times 4! = 864$

Sol.84. (c)

there are two type of words

CVCVCVCV or VCVCVCVC

(4! × 4!) + (4! × 4!) = 576 + 576 = 1152

Sol.85. (b)

all consonant together = 5! ×4! = 2880

Sol.86. (c)

if n is even number then ^nC_r will be greatest if we take r = n/2 so statement 1 is correct.

if n is odd number then ^nC_r will be greatest if we take r = (n+1)/2 or (n - 1)/2 so statement 2 is correct.

Sol.87. (c)

$^mC_2\ .\ {}^nC_2 = 60$

by hit and trial

if we take m = 2 then n cannot find

if we take m = 3 then n = 5

so m + n = 8

Sol.88. (c)

$x = \frac{12!}{2!.}$ and $y = \frac{12!}{2!.2!.2!.}$

x = 4y

Sol.89. (b)

total possible numbers = 4 × 4 × 3 × 2 = 96

number greater than 5000 will start from 5 so number of cases = 1 × 4 × 3 × 2 = 24

24 is the 25 % of 96

Sol.90. (b)

25! + 1 is an odd number than cannot be divisible by 26.

6! + 1 = 721 that is divisible by 7

so statement 1 is incorrect and statement 2 is correct.

Sol.91. (d)

number of 6 digit numbers ending with 2 or 4 = 4 × 4 × 3 × 2 × 1 × 2 = 192

number of 6 digit numbers ending with 0= 5 × 4 × 3 × 2 × 1 × 1 = 120

total = 192 + 120= 312

Sol.92. (c)

4! = 24 , which is divisible by 8

So 4! + 5! + 6! + …… will give zero remainder because all are individually divisible by 8

So remainder will obtained only by 0! + 1! + 2! + 3! = 10

that is 2

Sol.93. (d)

5! = 120 , which is divisible by 24

So 5! + 6! + 7! + …… will give zero remainder because all are individually divisible by 60

So remainder is 0! 1! + 2! + 3! + 4! = 34

Sol.94. (c)

5! = 120 , which is divisible by 24

So 5! + 7! + …… will give zero remainder because all are individually divisible by 24

So remainder is 1! + 3! = 7

Sol.95. (c)

Number of intersection points by 10 circles

$= 2\times{}^{10}C_2 = 90$

Sol.96. (b)

even digits 0,2,4,6,8

at first place zero cannot come so

number of ways = 4 × 5 × 5 × 5 = 500

Sol.97. (d)

for divisibility of 4 , last 2 digits should be divisible by 4

unit digit must be 2

if we take last two digit 12 then there is 2 possibilities 3512, 5312

if we take last two digit 32 then there is 2 possibilities 1532, 5132

if we take last two digit 52 then there is 2 possibilities 1352, 3152

so there will be 6 cases.

Sol.98. (d)

sum of number starting with 0 is 2220

sum of number starting with 1 is 7998

sum of number starting with 4 is 25332

sum of number starting with 5 is 31110

total = 64440

Sol.99. (c)

there are four possibilities

1st is man's relative 3 women and women's relative 3 men i.e. $^4C_0\,{}^3C_3\,{}^3C_3\,{}^4C_0$

2nd is man's relative 1 man and 2 women , women's relative 2 men and 1 woman i.e.$^4C_1\,{}^3C_2\,{}^4C_1\,{}^4C_0$

3^{rd} is man's relative 2 man and 1 women , women's relative 1 men and 2 woman
i.e. ${}^4C_2\,{}^3C_1{}^4C_2{}^3C_1$
4^{th} is man's relative 3 men and women's relative 3 women i.e. ${}^4C_0\,{}^3C_3{}^3C_3{}^4C_0$
${}^4C_0\,{}^3C_3{}^3C_3{}^4C_0 + {}^4C_1\,{}^3C_2{}^4C_1{}^4C_0 + {}^4C_2\,{}^3C_1{}^4C_2{}^3C_1 + {}^4C_0\,{}^3C_3{}^3C_3{}^4C_0 = 1 + 144 + 324 + 16 = 485$

Sol.100. (a)
there are total 12 points in the plane
so number of triangle formed by these points = ${}^{12}C_3 = 220$
if we take all three points of side PQ then triangle cannot formed. (1)
if we take all three points of side QR then triangle cannot formed (${}^4C_3$)
if we take all three points of side PR then triangle cannot formed (${}^5C_3$)
so number of triangles
$= 220 - 1 - {}^4C_3 - {}^5C_3 = 205$

Sol.101. (b)

2	26
2	13
2	6
2	3
	1

number of 2's = 13 + 6 + 3 + 1 = 23
i.e. $= 2^{23} = 4.2^{21} = 4.8^7$
so maximum value of k can be 7

Sol.102. (c)
max points of intersections of four straight lines $= {}^4C_2 = 6$
maximum intersection points of a circle and a straight line $= 2.\ {}^4C_1\ .\ {}^1C_1 = 8$
total points = 6 + 8 = 14

Sol.103. (b)
VCVCV
2 consonants can arrange as 2!
3 vowels can arrange as 3! / 2!
total required words = (3! × 2!) / 2! = 6

Sol.104. (c)
EQUATION
all vowels and consonants together
VVVVVCCC + CCCVVVVV
5! × 3! + 3! × 5! = 720 + 720 = 1440

Sol.105. (d)
a student choose (n – 2) courses out of n courses if 2 courses are compulsory
$= {}^{n-2}C_{n-4} = (n-2)(n-3)/2$

Sol.106. (a)
all four digits are odd and repetition is allowed = 5 × 5 × 5 × 5 = 625

Sol.107. (c)
Consonants C, P, T, L and Vowels A, I, A
Took all vowels together (CPTL)AIA,
So number of permutations =
$\Rightarrow \frac{4!}{2!.} \times 4! = 288$

Sol.108. (c)
Number of diagonals $= \frac{n(n-3)}{2} = 20$
$n^2 - 3n - 40 = 0$
n = 8

Sol.109. (c)
Consonants can arrange on three positions and vowels can arrange on 2 positions
So number of permutations =
$= 3! \times 2! = 12$
$n^2 - 3n - 40 = 0$
n = 8

Sol.110. (c)
x + y + z = 5
Take values 1, 1, 3 then no. of permutations $= \frac{3!}{2!} = 3$
Take value 2, 2, 1 then no. of permutations $= \frac{3!}{2!} = 3$
so total solutions = 6

Sol.111. (b)
Number of selections of r from $5n = {}^{5n}C_r$
Number of selections of n + r from $5n = {}^{5n}C_{n+r}$
Given that ${}^{5n}C_r = {}^{5n}C_{n+r}$
By formula ${}^{n}C_r = {}^{n}C_{n-r}$
5n = r + (n + r)
r = 2n

Sol.112. (d)
Number of selections of r from $5n = {}^{5n}C_r$
Number of selections of at most 3 things from 6 different things
$= {}^6C_0 + {}^6C_1 + {}^6C_2 + {}^6C_3 = 1 + 6 + 21 + 20 = 42$

Sol.113. (a)
$\frac{z}{y} = \frac{2x!}{x!} = 120$
Put x = 1, this value not satisfying
Put x = 2, this value also not satisfying
Put x = 3, this value satisfies above expression.
So (3x)! = 9! = 362880

Sol.114. (c)
We know that a pair of 5 and an even number make a zero at end,
Factorial of 5, 6, 7, 8, 9 have one zero at end of expansion because of only one 5 is there in the expansion.
Factorial of 10, 11, 12, 13, 14 have two zeros at end of expansion because of two times 5 are there in expansion.

Sol.115. (b)
Rule of Divisibility by 4: Last two digits should be divisible by 4
Take last 2 digit 12, than numbers are 4312,3412
Take last 2 digit 32, than numbers are 4132,1432
Take last 2 digit 24, than numbers are 1324,3124
So 6 numbers are possible which are divisible by 4.

Sol.116. (d)
Out of 8 points 4 are collinear , so number of triangles =
${}^8C_3 - {}^4C_3 = \frac{8\times7\times6}{3\times2\times1} - 4 = 56 - 4 = 52$

Sol.117. (c)
Out of 8 points 4 are collinear , so number of quadrilaterals =
${}^8C_4 - {}^4C_4 - {}^4C_3\,{}^4C_1 = \frac{8\times7\times6\times5}{4\times3\times2\times1} - 1 - 16$
= 70 – 1 – 16 = 53
Because if we take all 4 collinear points than quadrilateral is not possible, and if we take any 3 points from collinear points and one other point from non collinear points than triangle will form.

<u>You can access the solutions of PYQ through 6 Youtube video:</u>
1. Permutation and combination 2011 2012 | NDA mathematics previous year questions by Sandeep Brar
2. Permutation and combination 2013 2014 2015 | NDA mathematics previous year questions by Sandeep Brar
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